



THE NATIONAL INSTITUTE OF ENGINEERING

(An Autonomous Institution under VTU, Belagavi)

Manandavadi Road, Mysuru – 570008.

Department of Computer Science and Engineering



Course Plan for the term ODD 2026-27

Course Instructor: Sneha S

Course Title: Data Analytics and Visualization

L:T:P:TW/SL - 28:0:28:34

Term: Odd (2026 – 27)

Course Code:1BETC306

Sem and Sec: III sem and 'B' sec

LESSON PLAN

Session No.	Planned Date	Content	Actual Date/s	Mode of Delivery (Classroom/Online)	Link for online content	Link for Course Content	Comments, if any by CI/HoD
1.	10/08/2026	Module 1: Introduction to Data Analytics: Data Analytics: Definition, importance, and applications in industry		Classroom	https://youtu.be/37x5dKW-X5U?si=DuDcwIhpEM7yzwxS	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
2.	14/08/2026	Types of analytics: descriptive, diagnostic, predictive, prescriptive		Classroom	https://youtu.be/tIVXbHFnaVw?si=vUOhDtPcM5BtpPQ0	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
3.	17/08/2026	Data analytics lifecycle: problem definition, data collection, analysis, deployment		Classroom	https://youtu.be/4Cp6PkBKqX4?si=W7Ji5ljLVs8SBwUd	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	

4.	21/08/2026	Types of data: structured, semi-structured, unstructured; Time Series Data; Data sources and formats		Classroom	https://youtu.be/cYlzIeXAAAtI?si=CpoPu-Q6rr8DZOBA https://youtu.be/FsroWpkUuYI?si=NhE-ZSWjSMravisb	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
5.	24/08/2026	Python for data analytics: NumPy, Pandas, reading CSV/JSON files, indexing, filtering, grouping and aggregation.		Classroom	https://youtu.be/vtgDGrUiUKk?si=4xWwjgCdmaRV8oXR https://youtu.be/xECXZ3tyONo?si=68HFZA0MLjxm3NR	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
6.	28/08/2026	Overview of Big Data: characteristics (5Vs), ecosystem and tools		Classroom	https://youtu.be/s8V8nT26yKo?si=8-EieU340BTz5Pnc	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
7.	31/08/2026	Module 2: Exploratory Data Analysis (EDA): Descriptive statistics: mean, median, mode, variance, standard deviation, percentiles using pandas and Scipy		Classroom	https://youtu.be/SplCk-t1BeA?si=yxcSY9_6eVQF4N_r https://youtu.be/t4LOv9h-FJM?si=8NHkKxiWpV-cBEI	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	

8.	4/09/2026	Data distributions: normal, skewed, kurtosis; probability distributions		Classroom	https://youtu.be/HnMGKsupF8Q?si=3WLPVwpcy-roCujO	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
9.	7/09/2026	Correlation and covariance; Introduction to hypothesis testing basics (t-test, chi-square)		Classroom	https://youtu.be/uW0TapQ6UQU?si=ZIJDlokml07ogss https://youtu.be/YrhlQB3mQFI?si=ZW6FjJKH3N8vRGBP	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
10.	11/09/2026	Data visualization techniques: histograms, box plots, scatter plots, heatmaps		Classroom	https://youtu.be/csXmVBw8cdo?si=hD_j_4LWn97WQhEs https://youtu.be/ZlnJTGcPHQI?si=YQbiGXDktr4-T4aK	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
11.	18/09/2026	EDA using Pandas and Matplotlib/Seaborn libraries		Classroom	https://youtu.be/aYmcRnmZVGQ?si=nDjVxdNBZ8Aji3h3	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
12.	21/09/2026	Univariate, bivariate, and multivariate analysis		Classroom	https://youtu.be/JG8GRIMjp3c?si=fTdw9Yp8axwHKe1a	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	

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13.	5/10/2026	Module 3: Data Preprocessing and Feature Engineering: Data quality issues: missing values, duplicates, inconsistencies		Classroom	https://youtu.be/fGJk2S3Y4U?si=6TXLTihl8nFeJs2J	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
14.	5/10/2026	Handling missing data: imputation techniques		Classroom	https://youtu.be/-uC79UTOye8?si=CutSGN4C84o dlwiG	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
15.	9/10/2026	Outlier detection and treatment: IQR, Z-score methods		Classroom	https://youtu.be/A3gClkblXK8?si=TQNBOIrc3t mnptOi https://youtu.be/KFuEAGR3HS4?si=TLvKH0g1JDCT6a-C	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
16.	12/10/2026	Data Wrangling using Pandas: merging, joining, concatenation, grouping, and pivot tables		Classroom	https://youtu.be/TPivN7tpdwc?si=gTXm0_FjerqELnUy	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
17.	16/10/2026	Data transformation: normalization,			https://youtu.be/i3TWBQdoh9k?s	https://moodlegurukul.nie.ac.in/course/view.ph	

		standardization, encoding categorical variables		Classroom	i=I9ohstRb9Z9ziKhzhhttps://youtu.be/w2GglmYHfmM?si=H_zJonIn2q8CHCq6	p?id=7561#section-0	
18.	19/10/2026	Feature selection: filter, wrapper, and embedded methods; Introduction to Principal Component Analysis (PCA) for dimensionality reduction		Classroom	https://youtu.be/Sc-TNxW3PiI?si=vq7DaTwGIwhQ-pQR	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
19.	23/10/2026	Module 4: Predictive Analytics and Machine Learning Techniques: Introduction to Predictive Analytics and Machine Learning in Data Analytics.		Classroom	https://youtu.be/ukzFI9rgwfU?si=4ZE_lg0zT8HENF7v	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
20.	26/10/2026	Regression and Classification techniques for predictive analytics with model evaluation.		Classroom	https://youtu.be/SrZ6vifbmdc?si=p7o0XL15eeXJvF3B	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	

21.	30/10/2026	Unsupervised Learning Algorithms: K-Means Clustering, Hierarchical clustering		Classroom	https://youtu.be/J1cpZOYBi8A?si=ivNZTLam31r8CEi0 https://youtu.be/z8uvCejr5Qk?si=bWk01LktDRcjHCDK https://youtu.be/cdqN7_w6qKQ?si=BSEfhbKa6IryE7KZ	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
22.	2/11/2026	Model validation and evaluation metrics: Train/test split, cross-validation, overfitting/underfitting, accuracy, precision, recall, F1-score, confusion matrix.		Classroom	https://youtu.be/_CGTbkHwUHQ?si=nzPgvQZDaEezEh8W	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
23.	6/11/2026	Module 5: Data Visualization and Dashboard development with Case Studies: Principles of Data Visualization; colour theory and accessibility; data-ink ratio; identifying misleading charts		Classroom	https://youtu.be/Zs-Rbkr9LVI?si=SSeS_lFBvzewalvr https://youtu.be/JIMUzJzqaA8?si=jV-E0bnbxVzg_krW	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	

24.	9/11/2026	Visualization using Python: Matplotlib subplots and custom annotations		Classroom	https://youtu.be/GitXv_t0dZ4?si=tS7Rbe26VfhmZ9gL	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
25.	9/11/2026	Seaborn advanced plots. Plotly interactive charts		Classroom	https://youtu.be/UZDP9IPMqcs?si=h3zcYO0nBbYCqA1x	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
26.	13/11/2026	Interactive Dashboard Development with Streamlit: Building interactive web applications; file upload widget; sidebar filters and dropdowns		Classroom	https://youtu.be/6Eu2b34alsE?si=TZSNSu85PRX-gwk	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
27.	20/11/2026	Introduction to Tableau and Power BI: Interface overview; connecting to data sources; creating KPI metric cards, bar charts, and trend lines; applying filters and slicers; publishing and sharing a simple dashboard		Classroom	https://youtu.be/tNLp_JVipoQ?si=QMtiLiE4fB7-tVRz https://youtu.be/mmxVCFceQgU?si=32je9BUtcvD2Fvgj	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
28.	23/11/2026	Case studies: Healthcare Analytics, Retail Sales Analytics, Financial Analytics and Responsible Data Analytics		Classroom	https://youtu.be/Z9gq8DKexBo?si=97yMRwCiojRL2jWi	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	

Laboratory Session

Session No.	Planned Date/s for A1, A2, A3	Content	Actual Date/s for A1, A2, A3	Mode of Delivery (Classroom/ Lab/Online/ Virtual)	Link for online content/Virtual lab	Link for Course Content	Comments, if any by CI/HoD
1.		Introduction to Python for Data Analytics: NumPy and Pandas basics array operations, Data Frame creation, indexing, slicing.		Classroom	https://youtu.be/atG-l8VI8z4?si=cefkdmFaDGoM0HwK	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
2.		Exploratory Data Analysis: Loading datasets, computing descriptive statistics, identifying missing values and duplicates.		Classroom	https://youtu.be/FNLxYcUnow?si=RXiqNnaVhcUCqsYw	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
3.		Data Visualization using Matplotlib and Seaborn: histograms, box plots, scatter plots, pair plots, and correlation heatmaps.		Classroom	https://youtu.be/x0Ga1UG9szU?si=jjE3HE5fRtP5kwD1	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	

4.		Data Preprocessing: Handling missing data using imputation (mean, median, forward fill), and encoding categorical variables.		Classroom	https://youtu.be/aUnNWZorGmk?si=jORacAvPcSfiRhAb	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
5.		Outlier Detection and Treatment: Identifying outliers using IQR and Z-score methods and applying appropriate treatment.		Classroom	https://youtu.be/A3gClkbIXK8?si=ewmJdlQl4h9QOHIT https://youtu.be/tW4MMhkTzhs?si=6TIQPX8wkR-QdkP-	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
6.		Feature Engineering: Normalization, standardization, and Principal Component Analysis (PCA) for dimensionality reduction.		Classroom	https://youtu.be/Dv-Kk7PDEas?si=B8dCfiQOMWT9cKnj	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	
7.		Simple and Multiple Linear Regression: Building models, checking assumptions, and evaluating using R^2 and RMSE.		Classroom	https://youtu.be/a71-HNNAFw?si=BwIDJz0ZMID55Y5T https://youtu.be/SH4soR1O7fy?si=VwARpSn3ZtwvnI5	https://moodlegurukul.nie.ac.in/course/view.php?id=7561#section-0	

8.		Logistic Regression for Binary Classification: Model building, confusion matrix, precision, recall, F1-score evaluation.		Classroom	https://youtu.be/zM4VZR0px8E?si=4T6G9Q-lemqDiXNT	https://moodle.gurukul.nie.ac.in/course/view.php?id=7561#section-0	
9.		Decision Tree Classifier: Building, visualizing, and evaluating a decision tree model on a real-world dataset.		Classroom	https://youtu.be/PHxYNGo8Ncl?si=KDRpwk7KVLOfjKn5	https://moodle.gurukul.nie.ac.in/course/view.php?id=7561#section-0	
10.		K-Means Clustering: Applying K-Means on a dataset, elbow method to find optimal K, visualizing clusters.		Classroom	https://youtu.be/wW1tgWtkj4I?si=0cj78X28KtEF_QqG	https://moodle.gurukul.nie.ac.in/course/view.php?id=7561#section-0	
11.		Hierarchical Clustering: Building dendrogram and applying agglomerative clustering.		Classroom	https://youtu.be/YH0r47m0kFM?si=6SxxnD-TNMrdcDN2	https://moodle.gurukul.nie.ac.in/course/view.php?id=7561#section-0	
12.		Mini Project: End-to-end data analytics pipeline – data loading, EDA, preprocessing, model building, and visualization.		Classroom	https://youtu.be/4QkYy1wANXA?si=azG7JOCPIk3dOCfk	https://moodle.gurukul.nie.ac.in/course/view.php?id=7561#section-0	

Textbook: (Maximum of 2 books)

1. Jake VanderPlas, "Python Data Science Handbook: Essential Tools for Working with Data", O'Reilly Media, 2016.
2. Wes McKinney, "Python for Data Analysis", 3rd Edition, O'Reilly Media, 2022.

Reference Books:

1. Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow", 3rd Edition, O'Reilly Media, 2022.
2. Joel Grus, "Data Science from Scratch: First Principles with Python", 2nd Edition, O'Reilly Media, 2019.
3. Sebastian Raschka, "Python Machine Learning", 3rd Edition, Packt Publishing, 2019.
4. Andreas C. Müller and Sarah Guido, "Introduction to Machine Learning with Python", O'Reilly Media, 2016.
5. Vijay Kotu and Bala Deshpande, "Data Science: Concepts and Practice", 2nd Edition, Morgan Kaufmann, 2019.

Content beyond curriculum for Self-study along with link for course content:

Signature of Course instructor

Signature of Course Coordinator/Faculty Mentor

Signature of HoD

CIE	Total no. of teaching hours
Portions for TEST -1	12
Session 01 to Session 12	
Portions for TEST-2	12
Session 13 to Session 24	
Data Analytics & Exploration Series (DAES)	23
Session 05 to 28	
Experiential Learning	28
Integrated Lab	
SEE/SET	
20 marks from each Module with choice in two modules. Exam duration- 3 hours	

Evaluation Pattern

BT Level	Test-1 (25 Marks)	Test-2 (25 Marks)	Data Analytics & Exploration Series (DAES)	Experiential Learning (30 Marks)	SEE (100 Marks)
L1	-	-	-	-	-
L2	10	5	4	5	25
L3	10	10	6	5	35
L4	5	10	6	10	25
L5	-	-	-	10	15
L6	-	-	4	-	-

Data Analytics & Exploration Series (DAES) Evaluation Pattern

Criteria	Description	Marks
Problem Understanding	Understanding of the problem/scenario and identification of relevant analytical aspects	4
Analytical Approach	Application of analytics, preprocessing, visualization, AI tools, or analytical reasoning as appropriate	6
Insight Generation	Quality of findings, interpretations, conclusions, or recommendations	6
Communication & Presentation	Clarity, creativity, organization, and presentation of results	4
Total		20

Data Analytics & Exploration Series (DAES) Activities Name & Description

Sl. No.	Section	Activity Name	Brief Description	CO Mapping
1	A	Data Detective Challenge	Investigate data patterns, identify trends, anomalies, and possible causes from visualizations or datasets	CO1, CO2, CO3
2	B	Analytics Around Me	Explore analytics applications in daily life such as Amazon, YouTube, Maps, Swiggy, etc.	CO1, CO3
3	C	AI vs Human Analyst	Compare insights generated by AI tools with student-generated insights and validate results	CO1, CO2, CO3
4	D	Visualization Makeover Challenge	Identify misleading charts and redesign them using visualization principles	CO1, CO2, CO3, CO4
5	E	Personal Analytics Project	Collect and analyse personal data such as sleep, screen time, study hours, etc.	CO1, CO2, CO3, CO4
6	F	Insight Hunting Competition	Extract unique and valuable insights from a dataset and justify business relevance	CO2, CO3
7	G	Campus Analytics Observatory	Analyse institution-wide datasets and derive actionable recommendations	CO1, CO2, CO3, CO4
8	H	Analytics Myth Busters	Investigate and validate common analytics misconceptions using evidence and examples	CO1, CO3
9	I(AIML-A)	Data Visualization Meme Challenge	Create educational memes illustrating analytics and visualization concepts	CO1, CO4
10	J(AIML-B)	Build a Data Product	Develop a Streamlit, Tableau, or Power BI dashboard/application for a real-world problem	CO2, CO3, CO4

Evaluation Pattern

Component	Sub-Component	Evaluation Criteria	Marks
Lab Continuous Evaluation	Git Repository Lab Assessment (Programs 1–9)	Correctness of code, output accuracy, viva-voce understanding for each lab session	10
Lab Continuous Evaluation	Lab Test (End-Semester)	Independent execution of one assigned program from the syllabus within stipulated time; scaled down from 50 to 10 marks	10
Mini Project (Domain Case Study)	Problem Identification and Dataset Selection	Appropriateness of chosen domain (Healthcare/Retail/Finance) and dataset relevance to real-world problem	5
Mini Project (Domain Case Study)	Implementation — EDA, Model, Dashboard	Completeness of pipeline: data cleaning, minimum 3 EDA visualizations, one interactive Plotly/Streamlit/Power BI output	10
Mini Project (Domain Case Study)	Report and Presentation	Clarity of five data-driven findings, quality of written report, and oral presentation/demo	5
Suggested Tools/ AI assisted Tools: Python 3.x, Jupyter Notebook, Google Colab, Pandas, NumPy, SciPy, statsmodels, Matplotlib, Seaborn, Plotly, Streamlit, Tableau Public, Microsoft Power BI Desktop, Scikit-learn, GitHub.			

Signature of Course Instructor

Signature of Course Coordinator/Faculty Mentor

Signature of HoD